AI Glasses Carrier V1 - PCB Design Review

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Date	**2026-06-29**

- **Board:** ai_glasses_carrier_v1 **| CM4 SKU:** RM126-D4E32J0R35W28 (RK3576, 4GB LPDDR4X, 32GB eMMC, Wi-Fi6+BT5.x)
- **Pin-assignment status:** `REVIEW` (verified vs official pinout; not yet FROZEN)
- **Generated:** 2026-06-29 (from working sources; see footer)
- **Components:** 102 (81 P0) **| Passive families:** 22 (LOCKED 11 / VERIFY 6 / TUNE 5)
- **ERC:** 0 **| DRC:** 0 violations **| Footprints:** 87/102 real (15 VERIFY), 0 placed **| MPN set:** 27/102

> [!] **Stage:** schematic / pin-freeze. The PCB is intentionally an empty skeleton (outline + 6-layer stack + mounting holes + nets, **no footprints placed**). Per repo rule 7 this report makes **no** claim about layout/copper/clearance correctness - that is confirmed only in KiCad/3D after placement. Part values & specs are EVT candidates pending human review.

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1. Project status

Item	State
Pin assignment	[OK] 56 rows, all verified vs `radxa_cm4_pinout_v1.20.xlsx`
Freeze gate	[BLOCKED] `REVIEW` - blocked by 3 procurement/mechanical TBDs (not pins)
Schematic	[OK] element-level draft, net-connectivity verified
PCB layout	[ ] not started (skeleton only)
ERC / DRC	0 / 0 violations (see §2/§3)

2. Schematic status (ERC)

Schematic is generated from the BOM master (`scripts/carrier_bom.py`) as box symbols with global net labels; every net was checked with `kicad-cli sch export netlist`. The CM4 module is drawn as its three B2B connectors (J31=J3A, J32=J3B, J1) carrying the real CM4 pins.

ERC: 0 errors, 0 warnings - clean (`ai_context/erc.report.txt`: "ERC messages: 0 Errors 0 Warnings 0"). Confirms 102 symbols + 321 net labels with no unconnected-critical-pin, duplicate-net, or power-driver violations.

3. PCB status (DRC)

PCB = pre-layout skeleton: 70×55 mm outline (placeholder - carrier outline still a TBD), 6-layer stack (F.Cu / In1.GND / In2.PWR / In3.sig / In4.GND / B.Cu), 4× M2 mounting holes, declared nets, **no component footprints**. `Update PCB from Schematic` will populate it once footprints are assigned and the gate is FROZEN.

DRC: 0 errors, 0 warnings on the skeleton (`ai_context/drc.report.txt`). Meaningful DRC (clearances, diff-pair, copper) only applies after footprints are placed.

4. Footprint status

"Footprint" means three different things at three stages - they are NOT the same, and conflating them is what makes the status look contradictory:

Layer	Meaning	Status
(1) Footprint field (pointer)	Each symbol names *which* footprint to use	**102/102** assigned
(2) Real footprint exists/usable	That name maps to a real KiCad library footprint	**87/102** real, **15** still `VERIFY` placeholder
(3) Placed on the PCB	Physical footprint dropped into `.kicad_pcb`	**0** (pre-layout skeleton)

So: every symbol has a footprint *pointer* (needed for `Update PCB from Schematic`), **87** of those now point at real, datasheet-checked footprints from the installed KiCad 10 libraries (all passives + most connectors/ICs/test points), **15** are still placeholders, and **nothing is placed on the board yet**.

4.1 The 15 remaining placeholders - proposed real footprints

Each was researched against the installed KiCad libraries / the part datasheet. `custom` = must be drawn in `AI_Glasses.pretty` (no stock footprint); `confirm` = std-lib family identified, verify exact pads/EP vs the ordered part.

Ref	Part / package	Proposed footprint	Action
J31	Hirose DF40C-100DS-0.4V(51)	`AI_Glasses:Hirose_DF40C-100DS-0.4V_51 (from radxa-cm4-kicad / Hirose 2D)`	custom
J32	Hirose DF40C-100DS-0.4V(51)	`AI_Glasses:Hirose_DF40C-100DS-0.4V_51`	custom
J1	Hirose DF40C-100DS-0.4V(51)	`AI_Glasses:Hirose_DF40C-100DS-0.4V_51`	custom
U2	TPD4E05U06 USON-10 1.45x1.0 (DQA)	`Package_DFN_QFN:Texas_DSQ0010A_WSON-10-1EP_2x2mm (verify vs DQA)`	confirm
U16	TPD4E05U06 USON-10 1.45x1.0 (DQA)	`Package_DFN_QFN:Texas_DSQ0010A_WSON-10-1EP_2x2mm (verify vs DQA)`	confirm
U5	TPD2E009 (DRL=SOT-553 / DRY=SOT-23)	`Package_TO_SOT_SMD:SOT-23 or SOT-553 per ordered suffix`	confirm
U15	TPS25940 VQFN-20 (DNP/opt)	`Package_DFN_QFN:VQFN-20 3.5x4.5 (build/verify)`	confirm

Ref	Part / package	Proposed footprint	Action
U6	TPS61022 VQFN 1.5x2 (RWU)	`Package_DFN_QFN:QFN/SON per RWU 0.4mm pitch`	confirm
L1	Boost inductor 4.7uH, >=10A sat	`Inductor_SMD: 4x4mm class per chosen part`	confirm
U7	BQ25180 WSON-12 2x2	`Package_DFN_QFN:WSON-12 2x2 0.4mm (build/verify)`	confirm
U8	BQ29700 (DSE WSON-6 / SOT-23-5)	`Package per ordered suffix (WSON-6 or SOT-23-5)`	confirm
U9	MAX17048 TDFN-8 (2x3)	`Package_DFN_QFN:DFN-8-1EP_2x3mm_P0.5mm_EP0.61x2.2mm`	confirm
U10	ICM-42688-P LGA-14 2.5x3	`AI_Glasses:InvenSense_LGA-14_2.5x3mm`	custom
U14	u-blox MAX-M10S LGA-18 4.5x4.5	`AI_Glasses:ublox_MAX-M10S_LGA-18`	custom
D4	GNSS RF low-cap ESD 0.3pF	`Diode_SMD:D_0201_0603Metric or SOD-882 per part`	confirm

> The three CM4 B2B connectors (J31/J32/J1) and the GNSS module (U14) are the only hard `custom` items: the Hirose DF40C-100DS-0.4V(51) footprint + official CM4 STEP must be imported and the three connectors placed as one locked group (see §8 B2B fit-check).

5. Component list (by function)

CM4 Module / B2B Connectors (3)

Ref	Value	MPN	LCSC	Footprint	Function
J1	DF40C-100DS-0.4V(51)	DF40C-100DS-0.4V(51)	-	`AI_Glasses:VERIFY_B2B_J1_LOGICAL`	CM4 J1 (I/O): SAI1 audio, CSI3 camera MCLK, speaker enable
J31	DF40C-100DS-0.4V(51)	DF40C-100DS-0.4V(51)	-	`AI_Glasses:VERIFY_B2B_J31_LOGICAL`	CM4 J3A (low-speed): power, debug/GNSS UART, I2C, control GPIO

Ref	Value	MPN	LCSC	Footprint	Function
J32	DF40C-100DS-0.4V(51)	DF40C-100DS-0.4V(51)	-	`AI_Glasses:VERIFY_B2B_J32_LOGICAL`	CM4 J3B (high-speed): MIPI-CSI3 4-lane camera, USB2-OTG0, PDM1 mics

Camera (P0) (11)

Ref	Value	MPN	LCSC	Footprint	Function
C8	10µF	-	-	`Capacitor_SMD:C_0805_2012Metric`	Camera +3V3 bulk decoupling
C9	1µF	-	-	`Capacitor_SMD:C_0603_1608Metric`	Camera +3V3 decoupling
C10	100nF	-	-	`Capacitor_SMD:C_0603_1608Metric`	Camera +3V3 HF decoupling
C29	10µF	-	-	`Capacitor_SMD:C_0805_2012Metric`	Camera +5V bulk decoupling for Radxa Camera 4K
C30	100nF	-	-	`Capacitor_SMD:C_0603_1608Metric`	Camera +5V HF decoupling for Radxa Camera 4K
J2	FH35C-31S-0.3SHW(50)	FH35C-31S-0.3SHW(50)	C424662	`AI_Glasses:FH35C-31S-0.3SHW_50`	Camera FPC - Radxa Camera 4K / Sony IMX415-AAQR-C, 31-pin 0.3mm, 4-lane MIPI
R7	2.2k	-	-	`Resistor_SMD:R_0603_1608Metric`	Camera I2C SCL pull-up to +1V8 (cam OVDD=1.8V confirmed)
R8	2.2k	-	-	`Resistor_SMD:R_0603_1608Metric`	Camera I2C SDA pull-up to +1V8 (cam OVDD=1.8V confirmed)
U2	TPD4E05U06	TPD4E05U06	-	`AI_Glasses:TPD4E05U06_VERIFY`	MIPI-CSI low-capacitance ESD array for CSI3 clock + data lanes 0/1 (<0.4pF/line)

Ref	Value	MPN	LCSC	Footprint	Function
U3	LP5907-3.3 (low-noise LDO)	LP5907MFX-3.3	-	`Package_TO_SOT_SMD:SOT-23-5`	Camera 3.3V low-noise supply for Radxa Camera 4K, EN-gated
U16	TPD4E05U06	TPD4E05U06	-	`AI_Glasses:TPD4E05U06_VERIFY`	MIPI-CSI low-capacitance ESD array for CSI3 data lanes 2/3 (<0.4pF/line)

Audio (P0) (9)

Ref	Value	MPN	LCSC	Footprint	Function
C13	100nF	-	-	`Capacitor_SMD:C_0603_1608Metric`	Amp VDD decoupling
C23	1nF	-	-	`Capacitor_SMD:C_0603_1608Metric`	Class-D EMI cap (OUT+ to GND)
C24	1nF	-	-	`Capacitor_SMD:C_0603_1608Metric`	Class-D EMI cap (OUT- to GND)
FB1	Ferrite 600ohm@100MHz	-	-	`Inductor_SMD:L_0603_1608Metric`	Class-D speaker EMI bead (OUT+)
FB2	Ferrite 600ohm@100MHz	-	-	`Inductor_SMD:L_0603_1608Metric`	Class-D speaker EMI bead (OUT-)
J3	JST-SH 2p (spkr)	SM02B-SRSS-TB	-	`Connector_JST:JST_SH_BM02B-SRSS-TB_1x02-1MP_P1.00mm_Vertical`	Speaker connector (post-EMI); 2 wire-leads
MK1	SPH0641LU4H-1	SPH0641LU4H-1	-	`Sensor_Audio:Knowles_LGA-5_3.5x2.65mm`	PDM MEMS mic (L, SEL=GND)
MK2	SPH0641LU4H-1	SPH0641LU4H-1	-	`Sensor_Audio:Knowles_LGA-5_3.5x2.65mm`	PDM MEMS mic (R, SEL=VDD)
U4	MAX98357AETE+T	MAX98357AETE+T	C910544	`Package_DFN_QFN:QFN-16-1EP_3x3mm_P0.5mm_EP1.45x1.45mm`	Class-D I2S/SAI mono amp (TQFN-16, no MCLK)

USB-C (P0) (8)

Ref	Value	MPN	LCSC	Footprint	Function
C11	1µF	-	-	`Capacitor_SMD:C_0603_1608Metric`	VBUS input cap
D3	SMAJ5.0A (TVS)	SMAJ5.0A	-	`Diode_SMD:D_SMA`	USB VBUS ESD/TVS clamp (5V standoff)
J4	GCT USB4085 (USB2)	USB4085-GF-A	-	`Connector_USB:USB_C_Receptacle_GCT_USB4085`	USB-C receptacle (USB2 + 5V sink)
R1	5.1k	-	-	`Resistor_SMD:R_0603_1608Metric`	USB-C CC1 pull-down (Sink role)
R2	5.1k	-	-	`Resistor_SMD:R_0603_1608Metric`	USB-C CC2 pull-down (Sink role)
R3	0R / shunt	-	-	`Resistor_SMD:R_2512_6332Metric`	VBUS_PROT->5V series 0R/shunt (current log)
U5	TPD2E009	TPD2E009	-	`AI_Glasses:TPD2E009_VERIFY`	USB2 D+/D- ESD clamp
U15	TPS25940 (e-fuse, opt)	TPS25940	-	`AI_Glasses:VERIFY_4P`	VBUS e-fuse: OVP/OCP/reverse-block (DNP-able; bypass via R3 if unused)

Power (P0) (10)

Ref	Value	MPN	LCSC	Footprint	Function
C1	10µF	-	-	`Capacitor_SMD:C_0805_2012Metric`	+5V bulk decoupling
C2	100nF	-	-	`Capacitor_SMD:C_0603_1608Metric`	+5V HF decoupling
C3	10µF	-	-	`Capacitor_SMD:C_0805_2012Metric`	+3V3 bulk decoupling
C4	100nF	-	-	`Capacitor_SMD:C_0603_1608Metric`	+3V3 HF decoupling

Ref	Value	MPN	LCSC	Footprint	Function
C5	10µF	-	-	`Capacitor_SMD:C_0805_2012Metric`	+1V8 bulk decoupling
C6	100nF	-	-	`Capacitor_SMD:C_0603_1608Metric`	+1V8 HF decoupling
C21	100nF	-	-	`Capacitor_SMD:C_0603_1608Metric`	LDO U12 +3V3 output decoupling
C22	100nF	-	-	`Capacitor_SMD:C_0603_1608Metric`	LDO U13 +1V8 output decoupling
U12	TLV75733 (LDO)	TLV75733	-	`Package_TO_SOT_SMD:SOT-23-5`	3.3V rail regulator
U13	TLV75718 (LDO)	TLV75718	-	`Package_TO_SOT_SMD:SOT-23-5`	1.8V VCCIO rail regulator

Battery (P1) (14)

Ref	Value	MPN	LCSC	Footprint	Function
C7	22µF	-	-	`Capacitor_SMD:C_0805_2012Metric`	+VBAT bulk decoupling
C17	100nF	-	-	`Capacitor_SMD:C_0603_1608Metric`	Charger U7 VBUS decoupling
C18	100nF	-	-	`Capacitor_SMD:C_0603_1608Metric`	Fuel gauge U9 VDD decoupling
C19	100nF	-	-	`Capacitor_SMD:C_0603_1608Metric`	Boost U6 VIN decoupling
J5	JST-PH 3p	S3B-PH-SM4-TB	-	`Connector_JST:JST_PH_S3B-PH-SM4-TB_1x03-1MP_P2.00mm_Horizontal`	Battery connector (BAT+/BAT-/NTC)
L1	4.7µH	-	-	`AI_Glasses:POWER_INDUCTOR_7X7_VERIFY`	Boost inductor

Ref	Value	MPN	LCSC	Footprint	Function
R11	10k	-	-	`Resistor_SMD:R_0603_1608Metric`	Battery NTC bias (placeholder)
R12	2.2k	-	-	`Resistor_SMD:R_0603_1608Metric`	Charger/Fuel I2C6 SCL pull-up to +1V8
R13	2.2k	-	-	`Resistor_SMD:R_0603_1608Metric`	Charger/Fuel I2C6 SDA pull-up to +1V8
R20	100k	-	-	`Resistor_SMD:R_0603_1608Metric`	Boost EN pull-up to +VBAT; default boost enabled for bench bring-up
U6	TPS61022 (Boost)	TPS61022	-	`AI_Glasses:VERIFY_5P`	1S Li-Po -> 5V boost converter
U7	BQ25180	BQ25180	-	`AI_Glasses:VERIFY_8P`	1S Li-Po charger w/ power-path (I2C)
U8	BQ29700	BQ29700	-	`AI_Glasses:VERIFY_3P`	Single-cell battery protection (OVP/UVP/OCP)
U9	MAX17048	MAX17048	-	`AI_Glasses:VERIFY_6P`	I2C fuel gauge (SOC, low-batt IRQ)

IMU / Haptic (P1) (7)

Ref	Value	MPN	LCSC	Footprint	Function
C16	100nF	-	-	`Capacitor_SMD:C_0603_1608Metric`	IMU U10 VDD decoupling
C20	100nF	-	-	`Capacitor_SMD:C_0603_1608Metric`	Haptic U11 VDD decoupling
J6	JST-SH 2p	SM02B-SRSS-TB	-	`Connector_JST:JST_SH_BM02B-SRSS-TB_1x02-1MP_P1.00mm_Vertical`	Vibration motor connector
R9	2.2k	-	-	`Resistor_SMD:R_0603_1608Metric`	IMU I2C SCL pull-up to +1V8

Ref	Value	MPN	LCSC	Footprint	Function
R10	2.2k	-	-	`Resistor_SMD:R_0603_1608Metric`	IMU I2C SDA pull-up to +1V8
U10	ICM-42688-P	ICM-42688-P	-	`AI_Glasses:VERIFY_5P`	6-axis IMU (accel+gyro, I2C)
U11	DRV2605L	DRV2605L	-	`Package_SO:MSOP-10_3x3mm_P0.5mm`	LRA/ERM haptic driver

GNSS / Positioning (P0) (11)

Ref	Value	MPN	LCSC	Footprint	Function
C14	100nF	-	-	`Capacitor_SMD:C_0603_1608Metric`	GNSS VCC decoupling
C15	10µF	-	-	`Capacitor_SMD:C_0805_2012Metric`	GNSS VCC bulk decoupling
C25	shunt (DNP,tune)	-	-	`Capacitor_SMD:C_0402_1005Metric`	GNSS -match shunt, module side (DNP)
C26	shunt (DNP,tune)	-	-	`Capacitor_SMD:C_0402_1005Metric`	GNSS -match shunt, antenna side (DNP)
C27	DC-block (series,tune)	-	-	`Capacitor_SMD:C_0402_1005Metric`	GNSS RF series match / DC block
C28	100nF (active-ant)	-	-	`Capacitor_SMD:C_0603_1608Metric`	Antenna bias decoupling (DNP if passive)
D4	ESD low-cap RF 0.3pF	-	-	`AI_Glasses:RF_ESD_2P_VERIFY`	GNSS antenna ESD clamp
J8	U.FL / IPEX	U.FL-R-SMT-1	-	`Connector_Coaxial:U.FL_Hirose_U.FL-R-SMT-1_Vertical`	GNSS antenna connector (50ohm; external active/passive)
L2	RF choke (active-ant)	-	-	`Inductor_SMD:L_0402_1005Metric`	Bias-tee choke (DNP if passive antenna)
R17	10ohm (active-ant)	-	-	`Resistor_SMD:R_0603_1608Metric`	Active-antenna bias feed from +3V3 (DNP if passive)

Ref	Value	MPN	LCSC	Footprint	Function
U14	MAX-M10S-00B-01	MAX-M10S-00B-01	-	`AI_Glasses:VERIFY_9P`	u-blox GNSS (GPS/GLO/GAL/BDS), UART, VIO=1.8V

Indicators / Debug (5)

Ref	Value	MPN	LCSC	Footprint	Function
D1	LED Green	-	-	`LED_SMD:LED_0603_1608Metric`	Power LED (active-low GPIO sink)
D2	LED Blue	-	-	`LED_SMD:LED_0603_1608Metric`	Status LED (active-low GPIO sink)
J7	Header 1x4 2.54mm	-	-	`Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Vertical`	UART debug header (TX/RX/GND/VREF)
R4	1k	-	-	`Resistor_SMD:R_0603_1608Metric`	Power LED current limit
R5	1k	-	-	`Resistor_SMD:R_0603_1608Metric`	Status LED current limit

Test Points (16)

Ref	Value	MPN	LCSC	Footprint	Function
TP1	TestPoint 1.0mm	-	-	`TestPoint:TestPoint_Pad_D1.5mm`	Test point: VBAT
TP2	TestPoint 1.0mm	-	-	`TestPoint:TestPoint_Pad_D1.5mm`	Test point: 5V
TP3	TestPoint 1.0mm	-	-	`TestPoint:TestPoint_Pad_D1.5mm`	Test point: 3V3
TP4	TestPoint 1.0mm	-	-	`TestPoint:TestPoint_Pad_D1.5mm`	Test point: 1V8
TP5	TestPoint 1.0mm	-	-	`TestPoint:TestPoint_Pad_D1.5mm`	Test point: GND
TP6	TestPoint 1.0mm	-	-	`TestPoint:TestPoint_Pad_D1.5mm`	Test point: UART_TX
TP7	TestPoint 1.0mm	-	-	`TestPoint:TestPoint_Pad_D1.5mm`	Test point: UART_RX

Ref	Value	MPN	LCSC	Footprint	Function
TP8	TestPoint 1.0mm	-	-	`TestPoint:TestPoint_Pad_D1.5mm`	Test point: CAM_SCL
TP9	TestPoint 1.0mm	-	-	`TestPoint:TestPoint_Pad_D1.5mm`	Test point: CAM_SDA
TP10	TestPoint 1.0mm	-	-	`TestPoint:TestPoint_Pad_D1.5mm`	Test point: IMU_SCL
TP11	TestPoint 1.0mm	-	-	`TestPoint:TestPoint_Pad_D1.5mm`	Test point: IMU_SDA
TP12	TestPoint 1.0mm	-	-	`TestPoint:TestPoint_Pad_D1.5mm`	Test point: FUEL_SCL
TP13	TestPoint 1.0mm	-	-	`TestPoint:TestPoint_Pad_D1.5mm`	Test point: FUEL_SDA
TP14	TestPoint 1.0mm	-	-	`TestPoint:TestPoint_Pad_D1.5mm`	Test point: RESET
TP15	TestPoint 1.0mm	-	-	`TestPoint:TestPoint_Pad_D1.5mm`	Test point: RECOVERY
TP16	TestPoint 1.0mm	-	-	`TestPoint:TestPoint_Pad_D1.5mm`	Test point: CAM_EN

Buttons (8)

Ref	Value	MPN	LCSC	Footprint	Function
R14	10k	-	-	`Resistor_SMD:R_0603_1608Metric`	RECOVERY pull-up to +1V8
R15	10k	-	-	`Resistor_SMD:R_0603_1608Metric`	RESET pull-up to +1V8
R16	10k	-	-	`Resistor_SMD:R_0603_1608Metric`	POWER_KEY pull-up to +1V8
R18	100k	-	-	`Resistor_SMD:R_0603_1608Metric`	WL_nDIS pull-up to +1V8; default Wi-Fi enabled
R19	100k	-	-	`Resistor_SMD:R_0603_1608Metric`	BT_nDIS pull-up to +1V8; default Bluetooth enabled
SW1	Tact SW SMD	-	-	`Button_Switch_SMD:SW_Push_1P1T_NO_CK_KMR2`	RECOVERY / maskrom button (to GND)

Ref	Value	MPN	LCSC	Footprint	Function
SW2	Tact SW SMD	-	-	`Button_Switch_SMD:SW_Push_1P1T_NO_CK_KMR2`	RESET button (to GND)
SW3	Tact SW SMD	-	-	`Button_Switch_SMD:SW_Push_1P1T_NO_CK_KMR2`	POWER button (to GND)

6. CM4 pin assignment (verified vs official pinout)

Function	Signal	Conn	Pin	CM4 net	Voltage
Power	+5V_IN	J3A	77	5V_DCIN	5V
Power	+5V_IN	J3A	79	5V_DCIN	5V
Power	+5V_IN	J3A	81	5V_DCIN	5V
Power	+5V_IN	J3A	83	5V_DCIN	5V
Power	+5V_IN	J3A	85	5V_DCIN	5V
Power	+5V_IN	J3A	87	5V_DCIN	5V
Power	GND	J3A/J3B/J1	ALL_GND	GND	GND
Power	+3V3_OUT(CM4)	J3A	84,86	VCCIO6	3.3V
Power	+1V8_OUT(CM4)	J3A	88,90	VCC_1V8_S0	1.8V
Power	GPIO_VREF	J3A	78	GPIO_VREF	set=1.8V
Camera	MIPI_DPHY_CSI3_RX_CLKP	J3B	129	MIPI_DPHY_CSI3_RX_CLKP	diff_1V8 (100ohm)
Camera	MIPI_DPHY_CSI3_RX_CLKN	J3B	127	MIPI_DPHY_CSI3_RX_CLKN	diff_1V8 (100ohm)
Camera	MIPI_DPHY_CSI3_RX_D0P	J3B	117	MIPI_DPHY_CSI3_RX_D0P	diff_1V8 (100ohm)
Camera	MIPI_DPHY_CSI3_RX_D0N	J3B	115	MIPI_DPHY_CSI3_RX_D0N	diff_1V8 (100ohm)
Camera	MIPI_DPHY_CSI3_RX_D1P	J3B	123	MIPI_DPHY_CSI3_RX_D1P	diff_1V8 (100ohm)
Camera	MIPI_DPHY_CSI3_RX_D1N	J3B	121	MIPI_DPHY_CSI3_RX_D1N	diff_1V8 (100ohm)
Camera	MIPI_DPHY_CSI3_RX_D2P	J3B	135	MIPI_DPHY_CSI3_RX_D2P	diff_1V8 (100ohm)

Function	Signal	Conn	Pin	CM4 net	Voltage
Camera	MIPI_DPHY_CSI3_RX_D2N	J3B	133	MIPI_DPHY_CSI3_RX_D2N	diff_1V8 (100ohm)
Camera	MIPI_DPHY_CSI3_RX_D3P	J3B	141	MIPI_DPHY_CSI3_RX_D3P	diff_1V8 (100ohm)
Camera	MIPI_DPHY_CSI3_RX_D3N	J3B	139	MIPI_DPHY_CSI3_RX_D3N	diff_1V8 (100ohm)
Camera	I2C_CAM_SCL	J3A	80	I2C0_SCL_M1_TP	1.8V (PMUIO1)
Camera	I2C_CAM_SDA	J3A	82	I2C0_SDA_M1_TP	1.8V (PMUIO1)
Camera	CAM_RST_n	J3B	143	MIPI_CAM3_PDN_1V8	1.8V
Camera	CAM_MCLK	J1	59	MIPI_CSI3_CAM_CLKOUT_1V8	1.8V
PDM	PDM1_CLK1_M1	J3B	96	PDM1_CLK1_M1	1.8V (VCCIO_AUDIO)
PDM	PDM1_SDI1_M1	J3B	94	PDM1_SDI1_M1	1.8V (VCCIO_AUDIO)
PDM	PDM1_SDI2_M1	J3B	111	PDM1_SDI2_M1	1.8V (VCCIO_AUDIO)
SAI	SAI1_SCLK	J1	25	SAI1_SCLK_M0	1.8V (VCCIO_AUDIO)
SAI	SAI1_LRCK	J1	19	SAI1_LRCK_M0	1.8V (VCCIO_AUDIO)
SAI	SAI1_SDO0	J1	29	SAI1_SDO0_M0	1.8V (VCCIO_AUDIO)
SAI	SAI1_MCLK	J1	23	SAI1_MCLK_M0	1.8V (VCCIO_AUDIO)
Control	GPIO_SPKR_EN	J1	96	SPK_CTL_H	1.8V (VCCIO6)
USB	USB2_OTG0_DP	J3B	105	USB2_OTG0_DP	USB 3.3V analog (90ohm)
USB	USB2_OTG0_DM	J3B	103	USB2_OTG0_DM	USB 3.3V analog (90ohm)
UART	UART0_TX_M0_DEBUG	J3A	55	UART0_TX_M0_DEBUG	1.8V (VCCIO6)
UART	UART0_RX_M0_DEBUG	J3A	51	UART0_RX_M0_DEBUG	1.8V (VCCIO6)
GNSS	GNSS_UART_TX	J3A	47	UART7_TX_M0	1.8V (VCCIO6)
GNSS	GNSS_UART_RX	J3A	45	UART7_RX_M0	1.8V (VCCIO6)

Function	Signal	Conn	Pin	CM4 net	Voltage
GNSS	GNSS_PPS	J3A	40	SPI1_MISO_M0	1.8V (VCCIO6)
GNSS	GNSS_RST_n	J3A	44	SPI1_MOSI_M0	1.8V (VCCIO6)
Control	RECOVERY_n	J3A	93	MASKROM	1.8V
Control	RESET_n	J3A	92	RESET_L	1.8V
Control	POWER_KEY	J3A	99	PWRON_L	1.8V
Control	WL_nDIS	J3A	89	WL_NDIS	1.8V
Control	BT_nDIS	J3A	91	BT_NDIS	1.8V
Control	LED_PWR_n	J3A	95	NPWR_LED	1.8V (VCCIO6)
Control	LED_STATUS_n	J3A	21	PI_NLED_ACTIVITY	1.8V (VCCIO6)
I2C	I2C_IMU_SCL	J3A	56	I2C8_SCL_M1	1.8V (VCCIO6)
I2C	I2C_IMU_SDA	J3A	58	I2C8_SDA_M1	1.8V (VCCIO6)
Control	IMU_INT1	J3A	30	GPIO1_C1	1.8V (VCCIO6)
Control	VIB_PWM	J3A	48	PWM1_CH0_M2	1.8V (VCCIO6)
Control	VIB_EN	J3A	34	CAN1_TX_M3	1.8V (VCCIO6)
I2C	FUEL_I2C_SCL	J3A	35	I2C6_SCL_M3	1.8V (VCCIO6)
I2C	FUEL_I2C_SDA	J3A	36	I2C6_SDA_M3	1.8V (VCCIO6)
Control	CHG_INT_n	J3A	37	SPI1_CSN1_M0	1.8V (VCCIO6)
Control	LOW_BAT_INT_n	J3A	39	SPI1_CSN0_M0	1.8V (VCCIO6)

7. Passive component spec audit

Specs from `config/passive_bom_freeze.yaml`. **Verdict** checks good-practice compliance (MLCC voltage derating / DC-bias headroom, tolerance for critical positions). Status: LOCKED_CANDIDATE = ready for schematic-stage freeze (LCSC checked); PROCUREMENT_VERIFY = spec OK, confirm stock; TUNE_OR_EVT_SELECT = footprint/min-spec frozen, value/MPN by RF tuning or bench measurement.

7.1 Resistors - value, tolerance, package, tempco, power

Refs	Value	Role	Tol	Package	Tempco	Power	Temp °C	MPN	LCSC	Status
R7, R8, R9, R10, R12, R13	2.2k	1.8V I2C pull-up	1%	0603 / 1608 metric	100 ppm/C	0.1W	-55..155	RC0603FR-072K2L	C4190	LOCKED_CANDIDATE
R1, R2	5.1k	USB-C sink CC pull-down	1%	0603 / 1608 metric	100 ppm/C	0.1W	-55..155	RC0603FR-075K1L	C23186	LOCKED_CANDIDATE
R4, R5	1k	LED current limit	1%	0603 / 1608 metric	100 ppm/C	0.1W	-55..155	RC0603FR-071KL	C21190	LOCKED_CANDIDATE
R11, R14, R15, R16	10k	Reset/recovery/power pull-up and battery NTC placeholder bias	1%	0603 / 1608 metric	100 ppm/C	0.1W	-55..155	RC0603FR-0710KL	C25804	LOCKED_CANDIDATE
R18, R19, R20	100k	Wireless disable defaults and boost EN default	1%	0603 / 1608 metric	100 ppm/C	0.1W	-55..155	RC0603FR-07100KL	C25803	LOCKED_CANDIDATE
R17	10R	Active GNSS antenna bias feed, DNP if passive antenna	1%	0603 / 1608 metric	100 ppm/C	0.1W	-55..155	RC0603FR-0710RL	C22859	LOCKED_CANDIDATE
R3	0R / current shunt	Bring-up current measurement in +5V_SYS path	1% if populated as shunt; jumper option allowed for no-measure builds	2512 shunt footprint preferred; Kelvin sense pads recommended		1.0W	-55..155	TUNE_OR_EVT_SELECT	TUNE_OR_EVT_SELECT	TUNE_OR_EVT_SELECT

Resistor verdicts:

- [OK] **R7, R8, R9, R10, R12, R13** (2.2k): I2C pull-up; 2.2k valid for ~100-400 pF bus at <=400 kHz - confirm vs actual bus capacitance after layout.

- [OK] **R1, R2** (5.1k): USB-C CC sink pull-down - 1% locked (correct per spec).
- [OK] **R4, R5** (1k): LED current-limit - 1% fine (5% also OK).
- [OK] **R11, R14, R15, R16** (10k): Reset/recovery/power pull-up and battery NTC placeholder bias - tolerance 1%.
- [OK] **R18, R19, R20** (100k): Wireless disable defaults and boost EN default - tolerance 1%.
- [OK] **R17** (10R): Active GNSS antenna bias feed, DNP if passive antenna - tolerance 1%.
- [!] **R3** (0R / current shunt): 0ohm/shunt in the 5V high-current path - do NOT use a tiny 0603; select a real mohm shunt (0.005-0.02ohm, 2512, Kelvin) or a 1% jumper.

7.2 Capacitors - value, voltage, dielectric, package, temp

Refs	Value	Role	Dielectric	Voltage	Tol	Package	Temp °C	MPN	LCSC	Status
C2, C4, C6, C10, C13, C14, C16, C17, C18, C19, C20, C21, C22, C28, C30	100nF	Local HF decoupling and antenna-bias decoupling	X7R	50V	10%	0603 / 1608 metric	-55..125	CC0603KRX7R9BB104	C14663	LOCKED_CANDIDATE
C9, C11	1uF	Input/local decoupling	X7R	16V	10%	0603 / 1608 metric	-55..125	CL10B105KO8NNNC	C15849	LOCKED_CANDIDATE
C1, C3, C5, C8, C15, C29	10uF	Rail bulk decoupling	X5R	25V	10%	0805 / 2012 metric	-55..85	CL21A106KAYNNNE	C15850	LOCKED_CANDIDATE
C7	22uF	VBAT bulk decoupling	X5R	25V	20%	0805 / 2012 metric	-55..85	CL21A226MAQNNNE	C45783	LOCKED_CANDIDATE
C23, C24	1nF	Class-D speaker EMI shunt	X7R	50V	10%	0603 / 1608 metric	-55..125	CC0603KRX7R9BB102	C1669	LOCKED_CANDIDATE
C27	100pF initial / tune	GNSS RF DC block or series match	C0G/NP0	25V(min)	5% or better	0402 / 1005 metric	-55..125	TUNE_OR_EVT_SELECT	TUNE_OR_EVT_SELECT	TUNE_OR_EVT_SELECT

Refs	Value	Role	Dielectric	Voltage	Tol	Package	Temp °C	MPN	LCSC	Status
C25, C26	DNP / tune	GNSS pi-match shunt pads	C0G/NP0	25V(min)	5% or better	0402 / 1005 metric	-55..125	DNP_TUNE_KIT	DNP_TUNE_KIT	TUNE_OR_EVT_SELECT

Capacitor verdicts (DC-bias / derating vs rail):

- **C2, C4, C6, C10, C13, C14, C16, C17, C18, C19, C20, C21, C22, C28, C30** (100nF, X7R): [OK] 50V on +5V_SYS (5.0V) -> 10.0× margin, good DC-bias headroom
- **C9, C11** (1uF, X7R): [OK] 16V on USB_VBUS (5.0V) -> 3.2× margin, good DC-bias headroom
- **C1, C3, C5, C8, C15, C29** (10uF, X5R): [OK] 25V on +5V_SYS (5.0V) -> 5.0× margin, good DC-bias headroom - bulk MLCC: confirm effective C after DC bias (consider 0805/larger).
- **C7** (22uF, X5R): [OK] 25V on +VBAT (4.2V) -> 6.0× margin, good DC-bias headroom - bulk MLCC: confirm effective C after DC bias (consider 0805/larger).
- **C23, C24** (1nF, X7R): [OK] spec OK (no power rail mapped / EMI position).
- **C27** (100pF initial / tune, C0G/NP0): [OK] RF/timing position - C0G/NP0, 25V min; value RF-tuned.
- **C25, C26** (DNP / tune, C0G/NP0): [OK] RF/timing position - C0G/NP0, 25V min; value RF-tuned.

7.3 Magnetics, protection, indicators

Refs	Category	Value	Role	Package	Key ratings	MPN	LCSC	Status
L1	inductor	4.7uH	TPS61022-class 1S to 5V boost inductor	Shielded power inductor, about 7x7mm class	saturation_current_A_min=10; rms_current_A_min=6; dcr_mohm_max=25	TUNE_OR_EVT_SELECT	TUNE_OR_EVT_SELECT	TUNE_OR_EVT_SELECT
FB1, FB2	ferrite_bead	600R@100MHz	Class-D speaker output EMI bead	0603 / 1608 metric	impedance_ohm_at_100MHz=600; current_rating_A_min=1	BLM18AG601SN1D	PROCUREMENT_VERIFY	PROCUREMENT_VERIFY
L2	rf_inductor	RF choke / active antenna bias	GNSS active antenna bias tee, DNP if passive antenna	0402 / 1005 metric	inductance_initial=47nH initial / tune; q_requirement=High-Q at GNSS L1	TUNE_OR_EVT_SELECT	TUNE_OR_EVT_SELECT	TUNE_OR_EVT_SELECT
D3	tvs	SMAJ5.0A	USB VBUS surge/ESD clamp	SMA / DO-214AC	peak_pulse_power_W_min=400; vrwm_V=5	SMAJ5.0A	PROCUREMENT_VERIFY	PROCUREMENT_VERIFY

Refs	Category	Value	Role	Package	Key ratings	MPN	LCSC	Status
D4	esd	low-cap RF ESD, <=0.3pF	GNSS antenna RF ESD clamp	0402/DFN 1006 class	vrwm_V_min=5; capacitance_pF_max=0.3	PROCUREMENT_VERIFY_LOW_CAP_RF_ESD	PROCUREMENT_VERIFY	PROCUREMENT_VERIFY
U2, U16	esd_array	TPD4E05U06	MIPI CSI3 low-capacitance ESD arrays	USON/QFN class, verify footprint against chosen suffix	vrwm_V_min=5; capacitance_pF_max=0.4; channels=4	TPD4E05U06	PROCUREMENT_VERIFY	PROCUREMENT_VERIFY
U5	esd_array	TPD2E009	USB2 D+/D- ESD clamp	WSON/SOT package per selected suffix	vrwm_V_min=5; capacitance_pF_max=1.5; channels=2	TPD2E009	PROCUREMENT_VERIFY	PROCUREMENT_VERIFY
D1, D2	led	0603 LED, green/blue	Power/status indicators	0603 / 1608 metric	current_mA_nominal=1	PROCUREMENT_VERIFY_LED_COLOR	PROCUREMENT_VERIFY	PROCUREMENT_VERIFY

8. What's missing - open TBDs (freeze gate, §8.3)

ID	Resolved	Must confirm
`CM4_SKU`	[OK]	CONFIRMED RM126-D4E32J0R35W28: RK3576 / 4GB LPDDR4X / 32GB eMMC / Wi-Fi6+BT5.x. Antenna form tracked under ANTENNA.
`CAMERA_ELECTRICAL_PINOUT`	[OK]	RESOLVED: camera module is Radxa Camera 4K / Sony IMX415-AAQR-C; 31-pin FPC pinout captured in CM4_IMX415_design_files/Radxa_Camera_4K_31pin_pinout.csv and cross-referenced in docs/p0_p1_decisions_2026-06-28.md.
`CAMERA_CONNECTOR_MPN`	[OK]	RESOLVED: carrier J2 connector is Hirose FH35C-31S-0.3SHW(50), LCSC C424662. Footprint has been generated from the Hirose 2D drawing.
`CAMERA_CABLE`	[OK]	RESOLVED: use Radxa AC006, 31P 0.3mm to 31P 0.3mm, opposite-side FPC. Do not use AC008; AC008 is 31P to 15P.

ID	Resolved	Must confirm
`CAMERA_SCHEMATIC_GATE`	[OK]	RESOLVED for schematic stage: J2 pins 1-31, CSI3 4-lane mapping, P/N polarity, I2C/MCLK/RESET, voltage domains, power pins and NC pins are checked by scripts/audit_csi3_camera.py. RESET is active-low on project net CAM_RST_n; no PWDN pin is present in the Radxa Camera 4K 31-pin pinout.
`CAMERA_KICAD_IMPLEMENTATION`	[X]	PARTIAL: schematic source now uses 31-pin FH35C/Radxa Cam4K pinout on CM4 CSI3 4-lane, and the FH35C footprint has been regenerated from the Hirose 2D drawing. DEFERRED_TO_PRE_LAYOUT: AC006 physical validation, FPC contact side, FPC insertion direction, J2 Pin 1 physical check, 1:1 print, coupon test, and FPC bend/enclosure path. Official CM4 IO overlay radxa-cm4-io-radxa-camera-4k.dtbo is confirmed. Bring-up logs (dmesg/media/v4l2) remain open for EVT.
`AUDIO_PARTS`	[OK]	CONFIRMED: 2×SPH0641LU4H-1 mic, MAX98357AETE+T (C910544) amp, Ole Wolff OWS-091630W50A-8 8ohm/1W (C5840086) speaker
`GNSS_MODULE`	[OK]	CONFIRMED u-blox MAX-M10S-00B-01: UART, VCC=3.3V/VIO=1.8V, 50ohm U.FL, external active antenna (-match + bias-tee, DNP for passive)
`BATTERY`	[OK]	Reverse-calc: 1S 2-3×(35×18×4mm ~250mAh)=500-750mAh/1.85-2.8Wh. V1 bench use ~1.5-2Ah. >3-4h target needs AI duty-cycling (system-power issue, see report)
`POWER_5V`	[OK]	CONSERVATIVE FREEZE: keep TPS61022-class 1S->5V boost and design +5V_SYS for >=4A bench/boost margin. R3 remains the bring-up shunt/0R measurement point; final peak/thermal validation is deferred to bench testing.
`ANTENNA`	[OK]	CONSERVATIVE FREEZE: Wi-Fi/BT stays on the CM4 module/U.FL path; carrier does not add 2.4/5GHz RF routing. Layout must reserve the CM4 antenna keep-out and keep battery, metal hinge/screws, speaker magnet, and copper away from the antenna zone.

ID	Resolved	Must confirm
`BOARD_OUTLINE`	[X]	SOURCE GEOMETRY RESOLVED: CM4 module outline = 40.02x54.98mm from official DXF; CM4/IO Board DXF, STEP, placement map and dimensions drawing are the mechanical inputs. OPEN: project must freeze board length/width/radius, USB-C/camera/battery/debug exit direction, mounting holes, and battery-vs-CM4 position.
`B2B_CONNECTOR`	[X]	SOURCE FILES RESOLVED: carrier receptacle DF40C-100DS-0.4V(51), module plug DF40C-100DP-0.4V(51), 1.5mm mated height. OPEN: build 2-layer connector-only fit-check board and verify all three connectors press in together with no module lift/skew.
`PINMUX`	[OK]	PDM/SAI/I2C/UART/GPIO conflict map + Device Tree

Plus schematic-stage structural gaps: **no footprints placed on PCB** (skeleton), `VERIFY` placeholder footprints (§4), and the carrier board outline + B2B connector XY (needs official CM4 DXF/STEP aligned in KiCad).

9. Key recommendations

1. **USB-C CC (R1/R2, 5.1k):** locked at 1% [OK] - keep. Sink-role pull-downs to GND.
2. **I2C pull-ups (2.2k, R7-R13):** validate against measured bus capacitance after layout; 2.2k suits <=400 kHz / <=~400 pF. Camera I2C0, IMU I2C8, charger/fuel I2C6 are separate buses.
3. **5V-rail MLCC:** 10µF is 25V X5R / 1µF is 16V / 100nF is 50V -> all >=2× the 5V rail, good DC-bias margin. Still confirm **effective** capacitance of 10µF/0805 after DC bias on +5V_SYS.
4. **VBAT bulk (C7, 22µF/25V):** fine for 4.2V; high glasses-cell C-rate means keep/expand bulk near the boost input.
5. **R3 current shunt:** replace the 0ohm placeholder with a defined mohm shunt (2512, Kelvin) before measuring 5V power, or populate a 1% jumper for no-measure builds.
6. **GNSS RF (C25/C26/C27, L2):** C0G/NP0, values RF-tuned with the chosen antenna; C25/C26 stay DNP unless tuning needs shunt C; bias-tee (L2/R17/C28) DNP for a passive antenna.
7. **Boost inductor (L1):** select a >=10A-saturation shielded part sized to the measured 5V peak; footprint is a `VERIFY` placeholder.
8. **Replace all `VERIFY` footprints** (B2B, ESD arrays, RF ESD, inductor) with purchasable, library-verified footprints before layout.

Appendix A - full pin -> net connections

Ref	Pin	Net
J31	P77-87 +5V	+5V_SYS
J31	P78 VREF->1V8	+1V8
J31	GND	GND
J31	P55 DBG_TX	UART_DBG_TX
J31	P51 DBG_RX	UART_DBG_RX
J31	P47 GNSS_TX	GNSS_UART_TX
J31	P45 GNSS_RX	GNSS_UART_RX
J31	P40 GNSS_PPS	GNSS_PPS
J31	P44 GNSS_RST	GNSS_RST_n
J31	P80 CAM_SCL	CAM_I2C_SCL
J31	P82 CAM_SDA	CAM_I2C_SDA
J31	P56 IMU_SCL	IMU_I2C_SCL
J31	P58 IMU_SDA	IMU_I2C_SDA
J31	P35 FUEL_SCL	FUEL_I2C_SCL
J31	P36 FUEL_SDA	FUEL_I2C_SDA
J31	P30 IMU_INT	IMU_INT1
J31	P48 VIB_PWM	VIB_PWM
J31	P34 VIB_EN	VIB_EN
J31	P37 CHG_INT	CHG_INT_n
J31	P39 LBAT_INT	LOW_BAT_INT_n
J31	P95 LED_PWR	LED_PWR_n
J31	P21 LED_STAT	LED_STATUS_n
J31	P93 RECOVERY	RECOVERY_n
J31	P92 RESET	RESET_n
J31	P99 PWR_KEY	POWER_KEY
J31	P89 WL_DIS	WL_nDIS
J31	P91 BT_DIS	BT_nDIS
J32	P129 CSI3_CLKP	MIPI_CSI3_CLK_P
J32	P127 CSI3_CLKN	MIPI_CSI3_CLK_N
J32	P117 CSI3_D0P	MIPI_CSI3_D0_P
J32	P115 CSI3_D0N	MIPI_CSI3_D0_N

Ref	Pin	Net
J32	P123 CSI3_D1P	MIPI_CSI3_D1_P
J32	P121 CSI3_D1N	MIPI_CSI3_D1_N
J32	P135 CSI3_D2P	MIPI_CSI3_D2_P
J32	P133 CSI3_D2N	MIPI_CSI3_D2_N
J32	P141 CSI3_D3P	MIPI_CSI3_D3_P
J32	P139 CSI3_D3N	MIPI_CSI3_D3_N
J32	P143 CAM3_GPIO	CAM_RST_n
J32	P105 USB_DP	USB_DP
J32	P103 USB_DM	USB_DM
J32	P96 PDM_CLK	PDM1_CLK
J32	P94 PDM_D0	PDM1_DATA0
J32	P111 PDM_D1	PDM1_DATA1
J32	GND	GND
J1	P25 SAI_BCLK	SAI1_BCLK
J1	P19 SAI_LRCK	SAI1_LRCK
J1	P29 SAI_SDO	SAI1_SDO
J1	P23 SAI_MCLK_UNUSED	NC
J1	P59 CAM3_MCLK	CAM_MCLK
J1	P96 SPK_EN	SPKR_EN_n
J1	+1V8 (P88/90)	+1V8
J1	GND	GND
J2	1 GND	GND
J2	2 MDN4	MIPI_CSI3_D3_N
J2	3 MDP4	MIPI_CSI3_D3_P
J2	4 GND	GND
J2	5 MDN3	MIPI_CSI3_D2_N
J2	6 MDP3	MIPI_CSI3_D2_P
J2	7 GND	GND
J2	8 NC	NC
J2	9 NC	NC
J2	10 GND	GND

Ref	Pin	Net
J2	11 MDN2	MIPI_CSI3_D1_N
J2	12 MDP2	MIPI_CSI3_D1_P
J2	13 GND	GND
J2	14 MDN1	MIPI_CSI3_D0_N
J2	15 MDP1	MIPI_CSI3_D0_P
J2	16 GND	GND
J2	17 MCN	MIPI_CSI3_CLK_N
J2	18 MCP	MIPI_CSI3_CLK_P
J2	19 GND	GND
J2	20 MCLK	CAM_MCLK
J2	21 GND	GND
J2	22 NC	NC
J2	23 NC	NC
J2	24 SCL	CAM_I2C_SCL
J2	25 SDA	CAM_I2C_SDA
J2	26 NC	NC
J2	27 RESET_N	CAM_RST_n
J2	28 VCC3.3V	+CAM_3V3
J2	29 VCC3.3V	+CAM_3V3
J2	30 VCC5V	+5V_SYS
J2	31 VCC5V	+5V_SYS
U2	CLK_P	MIPI_CSI3_CLK_P
U2	CLK_N	MIPI_CSI3_CLK_N
U2	D0_P	MIPI_CSI3_D0_P
U2	D0_N	MIPI_CSI3_D0_N
U2	D1_P	MIPI_CSI3_D1_P
U2	D1_N	MIPI_CSI3_D1_N
U2	GND	GND
U16	D2_P	MIPI_CSI3_D2_P
U16	D2_N	MIPI_CSI3_D2_N
U16	D3_P	MIPI_CSI3_D3_P

Ref	Pin	Net
U16	D3_N	MIPI_CSI3_D3_N
U16	GND	GND
U3	VIN	+5V_SYS
U3	EN	CAM_PWR_EN
U3	GND	GND
U3	VOUT	+CAM_3V3
R7	1	+1V8
R7	2	CAM_I2C_SCL
R8	1	+1V8
R8	2	CAM_I2C_SDA
C8	1	+CAM_3V3
C8	2	GND
C9	1	+CAM_3V3
C9	2	GND
C10	1	+CAM_3V3
C10	2	GND
C29	1	+5V_SYS
C29	2	GND
C30	1	+5V_SYS
C30	2	GND
MK1	VDD	+1V8
MK1	CLK	PDM1_CLK
MK1	DATA	PDM1_DATA0
MK1	SEL	GND
MK1	GND	GND
MK2	VDD	+1V8
MK2	CLK	PDM1_CLK
MK2	DATA	PDM1_DATA1
MK2	SEL	+1V8
MK2	GND	GND
U4	VDD	+5V_SYS

Ref	Pin	Net
U4	GND	GND
U4	BCLK	SAI1_BCLK
U4	LRC	SAI1_LRCK
U4	DIN	SAI1_SDO
U4	SD	SPKR_EN_n
U4	GAIN	GND
U4	OUT+	SPK_OUT_P
U4	OUT-	SPK_OUT_N
FB1	1	SPK_OUT_P
FB1	2	SPK_P_F
FB2	1	SPK_OUT_N
FB2	2	SPK_N_F
C23	1	SPK_P_F
C23	2	GND
C24	1	SPK_N_F
C24	2	GND
J3	SPK+	SPK_P_F
J3	SPK-	SPK_N_F
C13	1	+5V_SYS
C13	2	GND
J4	VBUS	USB_VBUS
J4	GND	GND
J4	DP	USB_DP
J4	DM	USB_DM
J4	CC1	USB_CC1
J4	CC2	USB_CC2
J4	SHLD	GND
U5	DP	USB_DP
U5	DM	USB_DM
U5	GND	GND
R1	1	USB_CC1

Ref	Pin	Net
R1	2	GND
R2	1	USB_CC2
R2	2	GND
D3	1	USB_VBUS
D3	2	GND
U15	IN	USB_VBUS
U15	EN	USB_VBUS
U15	GND	GND
U15	OUT	VBUS_PROT
R3	1	VBUS_PROT
R3	2	+5V_SYS
C11	1	USB_VBUS
C11	2	GND
U12	VIN	+5V_SYS
U12	GND	GND
U12	VOUT	+3V3
U13	VIN	+3V3
U13	GND	GND
U13	VOUT	+1V8
C1	1	+5V_SYS
C1	2	GND
C2	1	+5V_SYS
C2	2	GND
C3	1	+3V3
C3	2	GND
C4	1	+3V3
C4	2	GND
C5	1	+1V8
C5	2	GND
C6	1	+1V8
C6	2	GND

Ref	Pin	Net
U6	VIN	+VBAT
U6	SW	BOOST_SW
U6	EN	BOOST_EN
U6	FB	+5V_SYS
U6	GND	GND
R20	1	+VBAT
R20	2	BOOST_EN
L1	1	+VBAT
L1	2	BOOST_SW
U7	IN	USB_VBUS
U7	SYS	+VBAT
U7	BAT	VBAT_P
U7	TS	BAT_NTC
U7	SCL	FUEL_I2C_SCL
U7	SDA	FUEL_I2C_SDA
U7	INT	CHG_INT_n
U7	GND	GND
U8	VDD	VBAT_P
U8	VM	VBAT_N
U8	OUT	+VBAT
U9	VDD	+VBAT
U9	CELL	+VBAT
U9	SCL	FUEL_I2C_SCL
U9	SDA	FUEL_I2C_SDA
U9	ALRT	LOW_BAT_INT_n
U9	GND	GND
J5	BAT+	VBAT_P
J5	BAT-	VBAT_N
J5	NTC	BAT_NTC
C7	1	+VBAT
C7	2	GND

Ref	Pin	Net
R11	1	BAT_NTC
R11	2	GND
U10	VDD	+1V8
U10	GND	GND
U10	SCL	IMU_I2C_SCL
U10	SDA	IMU_I2C_SDA
U10	INT1	IMU_INT1
R9	1	+1V8
R9	2	IMU_I2C_SCL
R10	1	+1V8
R10	2	IMU_I2C_SDA
U11	VDD	+5V_SYS
U11	GND	GND
U11	IN	VIB_PWM
U11	EN	VIB_EN
U11	OUT+	VIB_OUT_P
U11	OUT-	VIB_OUT_N
J6	M+	VIB_OUT_P
J6	M-	VIB_OUT_N
U14	VCC	+3V3
U14	V_IO	+1V8
U14	V_BCKP	+3V3
U14	GND	GND
U14	TXD	GNSS_UART_RX
U14	RXD	GNSS_UART_TX
U14	TIMEPULSE	GNSS_PPS
U14	RESET_N	GNSS_RST_n
U14	RF_IN	GNSS_RFIN
C27	1	GNSS_RFIN
C27	2	GNSS_ANT
C25	1	GNSS_RFIN

Ref	Pin	Net
C25	2	GND
C26	1	GNSS_ANT
C26	2	GND
L2	1	GNSS_ANT
L2	2	V_ANT_BIAS
R17	1	+3V3
R17	2	V_ANT_BIAS
C28	1	V_ANT_BIAS
C28	2	GND
J8	ANT	GNSS_ANT
J8	GND	GND
C14	1	+3V3
C14	2	GND
C15	1	+3V3
C15	2	GND
D1	A	LED_PWR_A
D1	K	LED_PWR_n
R4	1	+3V3
R4	2	LED_PWR_A
D2	A	LED_STAT_A
D2	K	LED_STATUS_n
R5	1	+3V3
R5	2	LED_STAT_A
J7	TX	UART_DBG_TX
J7	RX	UART_DBG_RX
J7	GND	GND
J7	VREF	+3V3
TP1	1	+VBAT
TP2	1	+5V_SYS
TP3	1	+3V3
TP4	1	+1V8

Ref	Pin	Net
TP5	1	GND
TP6	1	UART_DBG_TX
TP7	1	UART_DBG_RX
TP8	1	CAM_I2C_SCL
TP9	1	CAM_I2C_SDA
TP10	1	IMU_I2C_SCL
TP11	1	IMU_I2C_SDA
TP12	1	FUEL_I2C_SCL
TP13	1	FUEL_I2C_SDA
TP14	1	RESET_n
TP15	1	RECOVERY_n
TP16	1	CAM_PWR_EN
SW1	1	RECOVERY_n
SW1	2	GND
SW2	1	RESET_n
SW2	2	GND
SW3	1	POWER_KEY
SW3	2	GND
R14	1	+1V8
R14	2	RECOVERY_n
R15	1	+1V8
R15	2	RESET_n
R16	1	+1V8
R16	2	POWER_KEY
R18	1	+1V8
R18	2	WL_nDIS
R19	1	+1V8
R19	2	BT_nDIS
R12	1	+1V8
R12	2	FUEL_I2C_SCL
R13	1	+1V8

Ref	Pin	Net
R13	2	FUEL_I2C_SDA
C16	1	+1V8
C16	2	GND
C17	1	USB_VBUS
C17	2	GND
C18	1	+VBAT
C18	2	GND
C19	1	+VBAT
C19	2	GND
C20	1	+5V_SYS
C20	2	GND
C21	1	+3V3
C21	2	GND
C22	1	+1V8
C22	2	GND
D4	1	GNSS_ANT
D4	2	GND

Generated by `scripts/generate_design_review.py` from working sources (`carrier_bom.py`, `cm4_pinmap.py`, `config/passive_bom_freeze.yaml`, `ai_context/*.json`). Native KiCad/XLSX remain the source of truth (repo CLAUDE.md).